UNIVERSITY OF GHANA — DEPARTMENT OF COMPUTER SCIENCE
DCIT 208 — SOFTWARE ENGINEERING

D1: Project Proposal, Feasibility and Business Considerations

Team Name	SNART
Project Title	InspectoFleet
Client Name	Phanuel K. A. Wagba
Repository URL	https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-77
AI Assistance Disclosure	Claude (Anthropic) was used to help structure this proposal, organize sections per the deliverable instructions, draft the risk register and sprint plan template, and tighten wording. All client information, problem statements and scope decisions were supplied, reviewed, and approved by the team based on direct client conversation. See AI_USAGE.md for the full log.

A. Executive Summary

InspectoFleet is a digital vehicle inspection and fleet-tracking system being built this semester for Kal Car Rentals, a car rental business currently relying on a paper-based inspection process. Kal Car Rentals loses significant repair costs each month to disputes with customers over vehicle damage, because there is no photographic, time stamped evidence tied to rental contracts. The company also lacks real-time visibility in vehicle location and fuel/battery status, contributing to lost assets, unauthorized travel, and unbilled refueling charges.

This semester, the team will build a mobile-friendly digital inspection system that allows counter agents to capture time stamped photographic evidence of vehicle condition at handover and return, replacing the paper checklist process. The system will also include a fleet status dashboard giving fleet managers visibility into vehicle status and inspection history.

Task to be undertaken this semester

Design and build a working software prototype covering digital inspection workflow with photo/timestamp capture, inspection history and basic dispute evidence retrieval, and a fleet dashboard showing vehicle status based on inspection data.

Out of scope this semester

Integration with physical GPS/telemetry hardware is explicitly out of scope for this semester. The system will be designed so that GPS/telemetry data could be integrated in future (a defined data field and API endpoint for location/fuel data), but the team will not build, test, or rely on actual hardware integration. This decision was made to keep the semester's deliverable realistic and achievable, given the team's current experience level with hardware integration and the time available.

B. Client, Users and Intended Use

B1. Client

Client organization: Kal Car Rentals

Contact person / role: Phanuel K. A. Wagba, Director

B2. User Groups, Current Workflow and Pain Points

- Counter agents: currently complete paper checklists during vehicle handover/return; pain point: no photographic evidence, leads to disputes they cannot resolve.
- Fleet managers: currently have no real-time visibility into vehicle location or fuel/battery status; pain point: cannot proactively catch lost assets, unauthorized travel, or unbilled refueling.
- Kal Car Rentals (ownership/management): absorbs repair costs and disputes as a recurring expense; pain point: lack of centralized, trustworthy records.
- Customers: indirectly affected; pain point: disputes over damage responsibility, lack of transparent evidence at handover.

(Specific staff role titles and structure are based on the team's understanding from initial client conversation and will be refined as requirements are validated in D2.)

B3. Post-Course Usage

Kal Car Rentals intends to continue using the system after the course to manage vehicle inspections and reduce dispute-related losses. The team will hand off source code, documentation, and a deployment/operations guide so the client can maintain and extend the system.

B4. Three-Year Relevance

The core problem is the disputed vehicle damage and lack of fleet visibility, is not specific to current technology trends and will remain relevant as long as Kal Car Rentals operates a rental fleet. The system is designed with extensibility in mind, so it can grow with the business rather than needing to be replaced.

C. Overall Functionality and Preliminary Requirements

C1. High-Level Feature List by Role

Counter Agent

- Start a new inspection linked to a rental contract/vehicle
- Capture time-stamped photos of vehicle condition
- Record notes/damage observations against the inspection
- View past inspection history for a given vehicle

Fleet Manager

- View fleet-wide vehicle status (available, rented, under inspection)

- View inspection history and flagged damage across the fleet
- View live GPS location and fuel/battery level (out of scope this semester)

Management / Owner

- View summary reporting: disputes avoided, vehicles inspected, flagged damage trends

C2. Preliminary Functional Requirements

1. The system shall allow a counter agent to create a new inspection record linked to a specific vehicle and rental contract.
2. The system shall allow a counter agent to capture and attach time-stamped photographs to an inspection record.
3. The system shall store inspection records so they can be retrieved later as evidence in a dispute.
4. The system shall allow fleet managers to view the current status of each vehicle (available, rented, in for inspection).
5. The system shall allow authorized users to view inspection history for a given vehicle.
6. The system shall be designed with a defined data structure for future GPS/telemetry integration, without implementing live hardware integration this semester.

C3. Preliminary Non-Functional Requirements

- Usability: counter agents should be able to complete an inspection in a similar or shorter time than the current paper process.
- Reliability: inspection photo/timestamp data must not be lost once submitted.
- Security: only authorized staff accounts can create or view inspection records.
- Privacy: customer and vehicle data is treated as confidential.
- Performance: photo capture and upload should work reliably on typical mobile devices with average mobile data speeds in Ghana.
- Maintainability: code structured so a future developer can extend it.

C4. Major Assumptions and Constraints

- Counter agents will use a mobile device (smartphone) capable of running a modern mobile browser or lightweight app.
- Internet connectivity is available at the point of inspection (handover/return location), at least intermittently.
- The client does not yet have GPS/telemetry hardware installed in vehicles; this is a future investment, not assumed to be available this semester.
- Team has no prior production experience with mobile camera APIs or cloud photo storage

D. Main System Components

The system is decomposed into the following logical components. This decomposition keeps the photo-capture/evidence concern separate from fleet status reporting, since they have different reliability and update-frequency needs.

1. Mobile Inspection Interface

A mobile-friendly web interface used by counter agents to start inspections, capture photos, and record notes. Chosen as a web interface to avoid Appstore distribution overhead and simplify updates across devices.

2. Inspection & Evidence Service

Backend logic responsible for creating inspection records, storing photo evidence with timestamps, and linking inspections to vehicles and rental contracts. This is the core dispute-evidence component.

3. Fleet Status Dashboard

A web dashboard for fleet managers and ownership to view vehicle status, inspection history, and (in future) live location/fuel data. Kept as a separate component from the mobile interface since its users, devices, and update patterns differ.

4. Data Store

Central database holding vehicles, rental contracts, inspection records, and photo evidence references. Major data objects: Vehicle, RentalContract, InspectionRecord, EvidencePhoto, User/Staff.

5. (Future) Telemetry Integration Layer

Not built this semester. A defined interface/data contract will exist so that GPS/telemetry hardware data could be integrated later without redesigning the core system.

External systems: none confirmed yet beyond standard cloud photo storage. No payment or SMS integration is planned this semester.

E. Process and Milestone Plan

E1. Development Process

The team will follow an Agile approach with weekly sprints, aligned to the course's deliverable schedule (D5–D8 map directly to Sprint 1–Sprint 4). This fits the course structure directly, gives the team a regular cadence for client feedback, and matches the team's current skill level better than a heavier upfront-design process, given most members are new to structured software engineering practice.

E2. Sprint Plan (Week 6 – Week 12)

Week	Deliverable	Focus
Week 6	D3 — Backlog, UX Prototype, Sprint 1 Plan	Break SRS into estimated user stories; low-fidelity wireframes for inspection flow and fleet dashboard; plan Sprint 1
Week 7	D4 — Architecture, CI/CD Baseline	System architecture, data model, API contracts, CI pipeline, mid-sem defense

Week 8	D5 — Sprint 1 Review, Prototype v0.1	Core inspection workflow: login, start inspection, capture photos with timestamps
Week 9	D6 — Sprint 2 Review, Test Plan	Damage logging, inspection history, basic fleet vehicle list; formal test plan
Week 10	D7 — Sprint 3 Review, Security/Privacy/AI Audit	Access control, data privacy review, fleet status dashboard (manual/simulated data), observability
Week 11	D8 — Sprint 4 Review, Release Candidate	Polish, bug fixes, user manual draft, UAT prep with client
Week 12	D9 — Final Demo, Handoff Pack	Final demo video, full documentation, client handoff

E3. Visibility Plan

- Client (Kal Car Rentals): updated via WhatsApp/calls after each sprint review; a working demo or screenshots shared at each major milestone.
- Team: GitHub Issues and Project board kept current; weekly team call for status and task self-assignment.
- Lecturer: visibility through Sakai submissions and the GitHub repository itself (issues, PRs, commit history, release tags).

F. Business, Legal and Ethical Considerations

F1. Copyright / License Plan

The team and Kal Car Rentals will agree on a non-exclusive license allowing Kal Car Rentals to use, run, and modify the delivered system for its own business purposes after the course. The team retains the right to reference the project as coursework/portfolio evidence. This will be formalized in the Client Acceptance / Handoff Form (D9/D10).

F2. Open-Source Libraries

The team will use open-source libraries and frameworks where appropriate. Licenses will be reviewed to ensure they are compatible with both academic use and the client's business use.

F3. Data Protection and Confidentiality

Vehicle inspection photos and any customer/contract details are treated as confidential client data. Real client data will not be pasted into public AI tools or committed to the repository; sample/placeholder data will be used for development and testing where real data is not strictly necessary.

F4. NDA, Trade Secret, Patent or Conflict-of-Interest

No NDA, patent, or trade secret issue is currently anticipated. No team member has a conflict of interest with Kal Car Rentals beyond the standard client relationship being established for this course.

F5. No-Warranty Disclaimer

Kal Car Rentals will be informed at handoff that this is an academic project delivered without warranty and that the team's primary obligation is to the course's learning and assessment objectives alongside delivering a genuinely useful tool.

G. Risk Analysis

Top risks identified by the team, covering technical, client, team, and AI-related concerns:

#	Risk	Prob.	Impact	Owner	Mitigation
1	Client (Kal) becomes unresponsive or deprioritizes the project	Low	High	Product/Client Manager	Maintain weekly check-ins; escalate to lecturer if no contact for 1+ week
2	Scope creep — team attempts GPS/hardware integration despite it being out of scope	Medium	High	Architecture Lead	Explicit scope boundary documented here; revisit only after core MVP is stable
3	Team member non-participation / ghosting	Medium	Medium	All / Team Lead	Operating agreement consequence rules; reassign tasks early, document in minutes
4	Underestimating mobile photo-capture / offline-storage complexity	Medium	High	Implementation Lead(s)	Build and test photo capture early (Sprint 1); use proven libraries, avoid custom camera handling
5	Client data sensitivity (vehicle/customer photos, contract info) mishandled	Low	High	QA / Validation Lead	Classify data early in SRS; no real client data in public repo or public AI tools
6	Team lacks prior GitHub/CI experience, slowing delivery	High	Medium	DevOps / Release Lead	Use this semester's early deliverables to build proficiency; pair less experienced members with confident ones

7	Inconsistent or low-quality timestamped photo evidence (lighting, angle)	Medium	Medium	UX / Research Lead	Design guided photo-capture flow with on-screen prompts (e.g. 'capture front, rear, sides')
8	Device/OS fragmentation for counter agents' mobile devices	Medium	Medium	Architecture Lead	Target a single common platform (e.g. Android web app) rather than native multi-platform apps
9	Disputes about who approved/reviewed a PR, or unclear ownership	Low	Low	Documentation Lead	Enforce PR review/comment rule from operating agreement (B3/B4)
10	Final system not actually adopted by counter agents (resistance to change)	Medium	High	Product/Client Manager	Involve client/staff in UX review (D3); keep workflow close to existing paper process

G1. Client Availability Risk and Fallback

See D0 Section C5: if Kal Car Rentals becomes unresponsive before a future deadline, the team will pursue an alternative client within 48 hours and notify the instructor if none can be confirmed.

G2. AI Misuse Risk and Verification Plan

AI tools may be used to help structure documents, draft boilerplate code, or suggest test cases, but all AI-assisted output is reviewed, tested, and verified by a team member before being relied upon. All such use is logged in AI_USAGE.md per the course's AI contract (see D0 Section E).